Capstone Project Proposal: VeriCBAM (Stage 1 Core Summary)

Multimodal Evidence Fusion & Decision Support for CBAM Emissions Consistency Assessment

Student Name:	Cristhian David Cáceres Mateus (ID: 93515346)	**Institution:**	University of Europe for Applied Sciences (Potsdam)
Degree Program:	M.Sc. in Data Science (Winter 2026/2027)	**Faculty Supervisor:**	Dr. Humera Noor

1. Problem Statement & Regulatory Context

On 1 January 2026, the European Union's Carbon Border Adjustment Mechanism (CBAM, Regulation (EU) 2023/956) entered its definitive financial regime. Authorised CBAM Declarants importing carbon-intensive goods (Cement, Iron & Steel) must declare verified embedded emissions and surrender CBAM certificates priced to EU ETS allowances (Implementing Regulation (EU) 2025/2548). While Implementing Regulation (EU) 2025/2546 and Delegated Regulation (EU) 2025/2551 mandate accredited third-party verification and physical installation audits, auditing thousands of third-country plants creates a severe global bottleneck ("arithmetic of scarcity"), forcing unverified importers onto punitive default mark-ups (Implementing Regulation (EU) 2025/2621).

- Core Verification Challenge: Customs authorities and accredited verifiers lack an automated, physics-informed pre-screening tool to triage declarations against three primary evasion vectors before commissioning costly overseas site inspections: (A) Quantitative Understatement of fuel/throughput, (B) Stoichiometric Floor & Boundary Manipulation (including uncounted off-site blast furnace gas transfers), and (C) Process Route Misclassification (e.g., declaring primary integrated BF-BOF steel as low-carbon Scrap-EAF).

2. Core Research Question & Technical Hypotheses

Research Question: Can heterogeneous physical, orbital, historical, and co-pollutant evidence streams be systematically fused into an explainable decision-support model that screens declared CBAM emissions inconsistencies more reliably than isolated single-modality sources?

- H_1 (Stoichiometric Floor & Detection Blind Spot): Chemical mass-balance floors evaluated at conservative capacity utilization ($\eta_{min} = 0.50$) deterministically catch severe violations and route fraud, but exhibit a mathematical blind spot $\delta^* = 1 - (\eta_{min}/\eta_{actual})(EF_{floor}/EF_{typical})$ (≈ 55.6% for BF-BOF steel), necessitating statistical and observational fusion to catch moderate (10%–20%) understatements.
- H_2 (Orbital Trace-Gas Gate): Due to footprint dilution (5.5 × 3.5 km), wind advection, and winter cloud cover, Copernicus Sentinel-5P TROPOMI tropospheric NO_2 cannot directly meter annual CO_2 mass, but serves as an auxiliary operational activity gate when weighted by cloud fraction ($C_{sat} = 1 - f_{cloud}$).
- H_3 (Longitudinal & Co-Pollutant Fingerprinting): Leave-one-year-out combustion ratios (NO_x/CO_2) and one-sided longitudinal deviations (z_{drop}) detect selective CO_2 under-reporting without penalizing legitimate production growth.
- H_4 & H_5 (Dual-Stage Fusion & Abstention Gating): Combining all streams via confidence-gated Bayesian log-odds fusion (Model E1, abstaining when $C_{overall} < 0.50$) and facility-grouped logistic regression (Model E2) suppresses false alarms at strict audit budgets (FPR ≤ 5%, 10%).

3. Proposed Stage 1 Methodology & 5-Stream Multimodal Architecture

VeriCBAM functions strictly as an advisory screening copilot (preserving human verifier authority under EU law) that fuses five evidence streams:

1. Stream 1 — Chemical Stoichiometry & BAT Floors: Limestone calcination ($CaCO_3 \rightarrow CaO + CO_2$, 0.495 t CO_2/t clinker) plus EU CLM BREF minimum thermal fuel (0.275 t/t → $E_{floor,\ clinker} = 0.770$ t CO_2/t); iron ore reduction ($Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$) yielding $EF_{floor,\ BF\text{-}BOF} = 1.350$ t/t, $EF_{floor,\ DRI\text{-}EAF} = 0.550$ t/t, and $EF_{floor,\ Scrap\text{-}EAF} = 0.040$ t/t. Explicitly abstains (`stoich_applicable = False`) when installations export process gas (BFG/COG) across legal boundaries.
2. Stream 2 — Earth Observation Activity (Sentinel-5P TROPOMI): Level-2 tropospheric NO_2 contrast between a 5 km plant target disk and a 15–30 km background ring: $z_{sat} = (V_{target} - V_{bg}) / (1.4826 \cdot MAD_{bg})$, filtered at qa_value ≥ 0.50.
3. Stream 3 — Longitudinal Trajectory: One-sided log-ratio drop against leave-one-year-out facility median scaled by pooled MAD (σ_{pooled}): $z_{drop,\ i,\ t} = \max(0, -(\ln E_{declared,\ i,\ t} - M_{i,\ -t}) / \sigma_{pooled})$.
4. Stream 4 — Co-Pollutant Combustion Ratio: Leave-one-year-out deviation in permit-monitored NO_x / CO_2 mass ratios ($R_{i,\ -t}$).
5. Stream 5 — Technology & Route Verification: Cross-checking declared routes and nameplate capacities against the Global Energy Monitor (GEM) Steel & Cement Trackers and national Industrial Emissions Directive (IED) permits.

- Dual-Stage Decision Engines:

$$\text{logit } P(H \mid E) = \text{logit } P(H) + \Sigma_{k=1..5}\, w_k \cdot C_k \cdot \ln(LR_k), \qquad C_{overall} = ((\Sigma_{k=1..5}\, w_k\, C_k) / (\Sigma_{k=1..5}\, w_k))$$

- Model E1 (White-Box Bayesian Audit Dossier): Uses prior P(H)=0.08, domain weights (w in [1.0, 2.0]), 500 Monte Carlo weight-sensitivity draws, and an `Insufficient Evidence` gate ($C_{overall} < 0.50$) to generate traceable site-inspection dossiers.
- Model E2 (Supervised Triage Classifier): L_2-regularized Logistic Regression trained via 5-Fold `GroupKFold(groups=facility_id)` to rank-order customs declarations without cross-year facility identity leakage.

4. Empirical Reference Datasets & Validation Protocol

- Layer A (Real-World European Proxy Panel): Because overseas CBAM declarations are confidential, the project extracts 177 authentic facility-years (2018–2023) across 30 European installations (15 Cement, 15 Steel across 10 EU Member States) from the official EEA E-PRTR / IED v16.0 registry (`DOI: 10.2909/657ac3cb-affa-4295-a4a9-27b4f539adab`), paired with GEM capacity metadata.
- Layer B (325-Case Graded Benchmark): Evaluates six architectures (Model A Naive Prior, Model B Stoichiometry, Model C Sentinel-5P Alone, Model D Historical Alone, Model E1 Bayesian Fusion, Model E2 Logistic Triage) across 177 authentic compliant negatives, 140 graded synthetic understatements (δ in {5%, 10%, 20%, 30%, 50%}), and 8 route misclassifications.
- Validation Metrics: ROC-AUC, PR-AUC, Brier Score, Expected Calibration Error (ECE), Authentic FPR, Recall at matched false-alarm budgets (FPR = 5%, 10%), Auditor Triage Lift (Top 10%–30%), and 1,000 facility-clustered bootstrap 95% confidence intervals.

5. Core Deliverables & Key Regulatory References

- Deliverables: Modular Python package (`src/`), 8-tab interactive Streamlit Decision Cockpit (`src/ui/app.py`) with 1-click PDF/Markdown audit dossier export and zero-history supplier pre-check, automated reproduction pipeline (`scripts/reproduce_all.py`), and `pytest` unit test suite.
- Key References: Regulation (EU) 2023/956 (CBAM); Implementing Regulations (EU) 2025/2546 (Verification), 2025/2547 (Monitoring), 2025/2621 (Default Values); Delegated Regulation (EU) 2025/2551 (Accreditation); EEA E-PRTR v16 (2026); JRC CLM/IS BREF (2012–2013); Beirle et al. (2019), Science Advances (TROPOMI NO_2).